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I refactor legacy monoliths, fix forgotten architecture, improve performance, scalability, and reliability, and remove what doesn't belong. I'm flexible on tech stack — but to succeed, I need access to domain experts and decision makers.



Professional experience:

○ Independent Consultant [2024 May – present]

- Refactored a legacy data pipeline, fixing safety issues and enabling the implementation of new features within a reasonable amount of work..
- Refined the test suite to rely less on mockups and added full integration tests into the CI/CD process.
- Created multiple tools using Docker and Python to speed up development, testing and delivery

○ Software Architect - Isma Controlli [2019 February - 2024 April]

- Designed and led the development of an extendable framework for a [new family of free programmable BMS controllers](#) that has the potential to revolutionize BMS. Successfully used in prestigious buildings around Europe.
- Planned and oversaw development timelines, scope and execution protecting solution architecture from ad-hoc top down micromanagement.
- Led and coached a team in design patterns, clean code, and modular architecture fostering a culture of business focus, technical quality and productivity.
- Mentored a junior developer, initially assigned to me as an unskilled boot camp graduate, who within months progressed to become one of the company's top performers.
- Implemented most complex concepts, designed cost-efficient cloud communication strategies, designed CI pipelines for modular nugget based architecture and implemented various communication protocols.

Highlight: Took ownership of the product architecture for the company's strategic project soon after joining, ensuring its progress and stability during critical phases. My departure significantly slowed down the project's progress.

○ Senior Algorithms Programmer – Game Dev Ops [2021 June – 2021 December]

- Streamlined production by introducing best practices, including git-flow, scrum, and unit test.
- Developed algorithms for machine learning assisted automated UI-tests.
- Contributed to the migration of an on-premises Asp.Net solution to Azure, optimizing storage strategy for cost-efficiency combining Azure Blobs, Azure Files and SQLServer.
- Designed data models and interfaces for splitting monolithic code-base into modules. This significantly improved codebase maintainability and cleared the path for transforming modules into scalable microservices.

Highlight: When the machine learning team struggled due to lack of direction, I took the initiative to step in. Using my broad knowledge of computer science, I helped guide their efforts, bridged communication gaps, and made sure their work aligned with project goals, significantly boosting their productivity.

○ **Freelance Developer** [2011 April – 2019 February]

- Managed direct contacts with customers and executed a full development cycle of multiple applications from a concept to the market: native mobile clients for network applications, integrating various plugins.
- On the project for Nissan I was hired to develop application logic, but quickly adapted to writing shaders and post processes as that was most required. I learned these skills on the go and delivered within a tight, few-month timeline.
- On the project for Kenzo I developed custom video module in Unity3D as native one was too slow
- On the project for Playway I directed game production, managing concepts, design, art and business negotiations. Successfully delivered high-quality results within tight budget and schedule constraints overseeing consistency of the whole project.
- Live generation of 3D models, AI (scripted), data caching, audio and video streaming, Bluetooth, native sensors on mobiles, designing UI, asset optimization, technical documentation ...
- Transitioned from organisational assistance to programming role without any prior coding experience and ended preparing software architecture.

Highlight: Developed a video processing module. Shortly after that similar functionality was introduced in Messenger with orders of magnitude performance lag compared to my solution.

Technical exposure:

I don't memorize tools – I believe good engineering is primarily about understanding problems, not perfecting tools. I adapt to the challenge rather than trying to fit everything into what I already know. Here are a few of the things I've worked with:

- **Architecting Systems** - Modular, Scalable, SOLID, Extensible, Domain Driven Design, Distributed, Event Driven, Cloud, Dependency Injection.
- **Performance Optimization** – Multi-threading, Caching, Parallelization, Real-Time Processing, Data Optimization.
- **Automation & CI/CD** – Test-Driven Development, Integration Pipelines, Unit Tests, Integration Tests, Docker, Git, Jira, Confluence, Package Management.
- **Security & Reliability** – Ensuring data integrity, fault tolerance, and safety in critical systems that basically can never fail.
- **Programming Languages**: C# (>10 years), Swift (4 years), Java, Python, SQL, CG shader language, Objective-C...
- **Backend & Protocols** – REST, Modbus, BACnet, TCP, Bluetooth, Serial Port, SQL, HTTP(S), .Net, Entity Framework, Asp.Net, WebApi, Azure, RDB...
- **Mobile, Embedded & IoT** – Low Resources Environment, Real Time, Low Latency, FreeRTOS, Android, iOS, MetaWear, Bolts-Swift, CoreLocation, AVFoundation, nanoFramework, LittleFS...
- **Desktop and UI** - Windows, Unity3D, Mono, Winforms, WPF, UIKit, GPUImage, CoreGraphics, LiveCharts2, Video Processing, Audio Streaming...
- **Data & AI** – Regression, Clustering, Decision Trees, Neural Networks, Data Visualisation, Numpy, Pandas, Scipy, PYTorch, Tensorflow, matplotlib, scikit-learn...
- **Other**: Reflection, Json, CSV, XML, NUnit, XCTest, Proprietary Protocols, UI Testing...



Education:

- **Post Graduate Program in Cloud Computing** [present] - Drowning in the pricing of ephemeral data lakes in the managed virtualities of reserved containers.
- **Applied Data Science Program** [2024] - The most beautiful journey of mine in the last decade.
- **MA in Philosophy** at University of Gdańsk [2004-2009] - Studying raw entity in its void was good training for handling abstractions. It helps to avoid the pitfalls of stratospheric architects and allows navigation from the domain problem through neural network idea to parallelized machine code execution and, most importantly, to match technical and business ideas, which are often defined in different realms.
- **Technician of Electronics** at Technikum Łączności in Gdańsk [1997-2002] - I learned the magnificent beauty of the bare metal and understood how abstract code is expressed in the machine instructions and the flow of electrons.



Work Culture:

- **Outcome-oriented:** I don't care about ticking boxes in a to-do list. I aim to do what will provide value, even if that means questioning instructions or specifications, or cutting corners where it won't hurt too much. I reason from first principles to determine the best path forward—including economic factors but always keeping in mind alignment with long-term goals.
- **Devoted craftsman:** Good code is a thing of beauty and I want to live in a beautiful world. I desire to be proud of my job. My code shall be scalable, easy to read, easy for extensions, clean, efficient, focused. Following Rene Descartes method it's an art of making something complex looking trivially simple, so others can easily pick up, follow and build up.
- **Unfinished engineer:** I can always do better, I can always know better. I am an engineer, not a salesman – I look for weakness, possible improvements. Strengths are merely checkmarks indicating “no attention required”. This also applies to the usefulness. It's not perfect to spend big on something of small practical value. It's even less perfect to not deliver because time and budget run out.



Feedback:

- **Fellow programmers at GC5:** *We were wondering what we want to achieve in 10 years. Be as good as Adam, or be in some manager positions.*
- **Kenzo Mirror project manager:** *Adam is really good in his job, professional and available. He knows his job and my game doesn't have a single bug!*
- **Digital Arc System CEO:** *The entire team enjoyed working with Adam. [...] Adam stepped up during a time when the team was transitioning, and single-handedly pushed the project successfully forward even though a significant portion of the project appeared to be outside his comfort zone.*
- **Plangora CEO:** *I really hope I can work together with him again. Best guy I've ever worked with.*
- **Producer for Games at Shadow Factory:** *If I ever need someone to translate some unholy code into something readable and usable, I now know who to call.*
- **Insilico CEO:** *I loved working with Adam. [...] he stayed committed to it until the very end, and put his heart and soul into everything that we did and [...] we ended up with a product that I'm really proud of.*
- **Technical leader on the Nissan Juke project:** *Adam was responsible for core features, [...] The first task required writing custom shaders, dealing with performance and non-standard rendering for Oculus Rift [dev kit as it were beginnings]. The collaboration was very smooth, would happily work with Adam again.*